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40/100

CHEMICAL REACTION ENGINEERING

FINAL EXAM

CHE 321, SPRING 2025, CLOSED BOOK, TIME: 60 MIN

The endothermic gas phase reaction $A \rightarrow B + C$ is carried out in a single steam-jacketed fluidized bed reactor, where the temperature of the steam circulating through the jacket remains constant at $T_a = 300^\circ\text{C}$.

For the following data, calculate the steady-state reactor temperature.

Conversion: $X = 85\%$

Heat-transfer area of the jacket: $A = 5 \text{ m}^2$

Overall heat transfer coefficient of the jacket: $U = 500 \frac{\text{J}}{\text{s.m}^2.\text{K}}$

Heat of reaction per unit mole of species A at 25°C : $\Delta_r H = 120 \frac{\text{kJ}}{\text{mol}}$

Feed temperature: $T_0 = 40^\circ\text{C}$

Feed molar flow rates: $F_{A0} = 10 \frac{\text{mol}}{\text{s}}$, $F_{I0} = 2 \frac{\text{mol}}{\text{s}}$ (inert), $F_{B0} = F_{C0} = 0$

Heat capacities (assume constant) $c_{pA} = 90 \frac{\text{J}}{\text{mol.K}}$, $c_{pB} = 60 \frac{\text{J}}{\text{mol.K}}$, $c_{pC} = 40 \frac{\text{J}}{\text{mol.K}}$, $c_{pI} = 50 \frac{\text{J}}{\text{mol.K}}$

Reactor energy balance: $\frac{dU}{dt} = \sum_{\alpha} F_{\alpha 0} h_{\alpha}(T_0) - \sum_{\alpha} F_{\alpha} h_{\alpha}(T) + \dot{Q} + \dot{W}_s$

Conversion $X \equiv \frac{F_{A0} - F_A}{F_{A0}}$

$$\sum F_{\alpha} h_{\alpha}(T_0) - \sum F_{\alpha} h_{\alpha}(T) + \dot{Q} + \dot{W}_s = 0$$

$$-r_A = k C_A = k C_{A0} \frac{F_A}{F_T} \left(\frac{T_0}{T} \right)$$

$$F_{A0} - F_A + r_A V = 0$$

$$\frac{dF_A}{dV} h_A + \frac{dh_A}{dV} F_A + UA(T_a - T_0)$$

$$C_A = C_{T0} \frac{F_{A0}}{F_T} \left(\frac{T_0}{T} \right)$$

$$\frac{dT}{dV} = \frac{\sum \frac{dF_{\alpha}}{dV} h_{\alpha}}{\sum C_{p\alpha} F_{\alpha}} - UA(T_a - T_0)$$

$$F_A = F_{A0} \left(\frac{T_0}{T} \right)$$

$$0.85 = \frac{10 - F_A}{10}$$

$$C_p = -40 \text{ J/mol.K}$$

$$8.5 = 10 - F_A$$

$$F_A = 1.5 \text{ mol/s}$$

$$T_a = 300^\circ\text{C} \rightarrow 573\text{K}$$

$$T_0 = 40^\circ\text{C} \rightarrow 313\text{K}$$

$$506(5)(573-313) = 650,000 \text{ J/s}$$

$$\frac{df_A}{dV} = r_{AV}$$

$$F_{A0} \frac{dx}{dV} + \frac{dh_A}{dV} f_A + UA(T_a - T_o)$$

$$\frac{dT}{dV} = \frac{\sum F_{A0} \frac{dx}{dV} h_A}{\sum C_{p,A} h_A} - UA(T_a - T_o)$$

$$dT = \frac{\sum F_{A0} h_A}{\sum C_{p,A} h_A} dx$$

$$\int_{313}^T dT = \int_0^{.85} \frac{10(120)}{90(120) + 60(120) + 40(120) + 50(120)} dx$$

$$T - 313 = .0354(6)$$

$$T = 314.1 \text{ K}$$

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